Thomas Verelst

PhD Researcher in Computer Vision - ESAT-PSI, KU Leuven

@ thomasverelst@outlook.com +32471396885 Leuven, Belgium thomasverelst.github.io linkedin.com/in/tverelst

Reducing inference time of deep learning by writing extra code for model compression and conditional computation.
Strong coding skills, endless curiosity and a touch of entrepreneurship.

EDUCATION

PhD candidate in Engineering Science

Computer vision and deep learning

Oct 2018 - Present KU Leuven, Belgium

- PhD promotor: Prof. Tinne Tuytelaars
- Efficient convolutional neural networks (CNN), model compression and dynamic neural networks with conditional computation, reducing the computational cost of state of the art networks for classification, segmentation, human/hand pose estimation, and object detection.
- Implementation of computer vision algorithms for industry partners
- Teaching of courses and master theses.

Master of Science: Electrical Engineering

Embedded systems and multimedia

Sept 2016 – June 2018 KU Leuven, Belgium

- Magna Cum Laude (81.28%)
- with one-year exchange to EPFL Lausanne, Switzerland

Bachelor of Science

Electrical engineering - computer science

Sept 2013 – June 2016 KU Leuven, Belgium

- Magna Cum Laude (82.67%)

Secondary school

Latin-Mathematics

Sint-Jan Berchmanscollege Westmalle, Belgium

LEAD AUTHOR PUBLICATIONS

SegBlocks: Towards Block-Based Adaptive Resolution Networks for Fast Segmentation

- Thomas Verelst and Tinne Tuytelaars, ECCV Embedded Vision Workshop 2020 (Best Short Paper Award)
- Reduces computational cost and memory usage of deep neural networks for Cityscapes semantic segmentation by adapting the processing resolution of image regions.

Dynamic Convolutions: Exploiting Spatial Sparsity for Faster Inference

- Thomas Verelst and Tinne Tuytelaars, CVPR 2020
- Reduces computational cost of CNNs by selectively applying convolutional operations using end-to-end trained attention, on ResNet, MobileNetV2 and Stacked Hourglass architectures.
- `https://arxiv.org/abs/1912.03203`

Generating Superpixels using Deep Representations

- Thomas Verelst, Matthew Blaschko and Maxim Berman
- Master thesis work (grade 17.9/20). Extended abstract in DeepVision workshop, CVPR 2018.
- `https://arxiv.org/abs/1903.04586`

INTERNSHIPS AND SUMMER SCHOOLS

AI/ML R&D Internship (PhD) at Apple

Data-efficient deep learning

April - August 2021 Zürich, Switzerland

ICVSS 2019 summer school

Focusing on computer vision and deep learning

July 2019 Sicily, Italy

Internship (master) at Nokia Networks

6 weeks, Summer 2017 Antwerp, Belgium

Internship (bachelor) at Semicon Sp. z o.o.

6 weeks, Summer 2016 Warsaw, Poland

SKILLS

Languages

Dutch: native	French: fair
English: fluent	German: basic

Programming

- Proficient: Python, PyTorch, Numpy, OpenCV, MATLAB, GIT
- Familiar: C/C++, Bash, CUDA, Scikit-learn, Java
- Beginner: Tensorflow, Keras, .NET C#
- Electronics: basic VHDL, PCB design (Altium), FPGA, microcontrollers
- WebDev: HTML5, PHP, CSS3, Javascript, MySQL, jQuery, Wordpress, Flask, Bootstrap

EXTRACURRICULAR

Avid participant in tech events such as hackathons and workshops. When not programming, I am probably hiking, photographing or skiing in the Alps.

- **Hackathons**: building prototypes in 48 hours
  - HackJunction2020 Connected: challenge winner - smart shopping assistant app
  - HackZurich 2018 (Europe's largest hackathon): finalist and IBM challenge winner - risk forecasting app
  - HackTheAlps2018: public award and challenge winner - ski area estimation from aerial imaging
- **AFC Participant and Student Consultant** (2015-2017): IT consulting in a multi-disciplinary team of 5. Leadership, presentation & productivity workshop tracks.
- **Full-stack web developer** (2012-...): development and support of a templating framework. PageRank 5 with 500.000 visitors since 2012. `http://metro-webdesign.info/`

REFERENCES

PhD promotor: Prof. Tinne Tuytelaars
`tinne.tuytelaars@esat.kuleuven.be`